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POE PART 2

Table of Contents

Overview

Prototype of a full-fledged e-commerce web application designed for The IIE POE (Proof of Event). Users can view clothing products, register, login, and purchase. Administrators will have the ability to manage users.

Features

Customer Features

- User Registration - Register new user with validation (all fields are mandatory)

- User Login- Authenticate the user with password hashed by MD5 and sticky forms

- View Products - View list of clothing products with image, price, and size options

- Shopping Cart - Add/remove product before making an order

- View Orders - View order history with status

- Manage Account - View account details

Admin Features

- Admin Login - Separate login page for admin users

- Verify New User - Verify customer registration (login is blocked until verification)

- Manage Users- CRUD operations for managing customer details

- Analytics Dashboard - View number of pending verifications and total customer count

Technical Features

- Sticky Forms - Persist user input in

Technology Stack

Component	Technology
Backend	PHP 7.4+ (MySQLi extension)
Database	MySQL 5.7+ / MariaDB
Frontend	HTML5, CSS3, JavaScript
Server	Apache (XAMPP/WAMP/MAMP)
Database Tool	phpMyAdmin

Installation Guide

Prerequisites

1. Install Local Server Environment

- [WAMP](https://www.wampserver.com/) (for Windows)

2. Ensure requirements:

- PHP 7.4 or above
- MySQL 5.7 or above
- Apache web server

Installation Steps

Step 1: Download & Extract

Clone the project

git clone https://github.com/yourusername/closet-connect.git

OR unzip the folder into your web server directory

Step 2: Place Files Into Web Server Folder

Server	Directory Location
XAMPP	C:\xampp\htdocs\closet-connect\
WAMP	C:\wamp64\www\closet-connect\
MAMP	/Applications/MAMP/htdocs/closet-connect/

Step 3: Start Server Services

XAMPP Control Panel

• Click "Start" next to Apache

• Click "Start" next to MySQL

Step 4: Modify the database connectivity file

DBConn.php found under includes directory

Modify with your database credentials as follows:

php

```
<?php
$servername = "localhost";
$username = "root";   // MySQL user name
```

Database Setup

METHOD 1 – With phpMyAdmin (Recommended)

Step 1: Access phpMyAdmin via URL: http://localhost/phpmyadmin

Username: root

Leave password empty for XAMPP

Step 2: Import Database

a. Go to "New Database" and create ClothingStore database

b. Select database

c. On left pane, go to "Import"

d. Choose file: myClothingStore.sql

e. Click "Go"

METHOD 2 - Automatically using PHP script

PHP setup script can be accessed at:

http://localhost/closet-connect/sql/loadClothingStore.php

This script will:

- Delete database if it already exists
- Create a new database named ClothingStore
- Then create tables (tblUser, tblAdmin, tblClothes, tblOrder)
- Populate the tables with 30 entries each.

METHOD 3 - Import user information from text file

To import users from the userData.txt file:

http://localhost/closet-connect/sql/createTable.php

Default Login Credentials

Admin User

Parameter =Values

Username =admin

Password =admin123

Customer Users

Username	Email Address	Password
john_doe	john.doe@example.com	password123
jane_smith	jane.smith@example.com	password123
mike_brown	mike.brown@example.com	password123

File Structure

```
Pastimes/
├── index.php            # Main landing page
├── login.php            # Customer login (sticky form)
├── admin_login.php      # Admin login portal
├── register.php         # User registration
├── account.php          # User dashboard
├── admin_dashboard.php  # Admin CRUD interface
├── cart.php             # Shopping cart
├── logout.php           # Session destroy
├── userData.txt         # Sample user data file
├── README.md            # This file
├── includes/
│   └── DBConn.php       # Database connection
├── css/
│   └── style.css        # Styling
├── sql/
│   ├── database.sql     # Database schema
│   ├── createTable.php  # Recreate tblUser from text file
│   ├── loadClothingStore.php # Complete DB setup
│   └── myClothingStore.sql  # Full export with 30 entries
└── assets/              # Images and static assets
```

Usage Guide

For Customers

1. Register New Account

1. Click "Register" in navigation
2. Fill all required fields (no empty fields allowed)
3. Click "Register"
4. Wait for admin verification (cannot login immediately)

2. Login to Account

1. Click "Customer Login"
2. Enter username and email
3. Enter password
4. If successful: See "User John Doe is logged in" message
5. If failed: Form retains entered data (sticky form)

3. Browse and Shop

1. Browse products on homepage
2. Click "Add to Cart" on desired items
3. View cart by clicking cart icon or navigating to cart
4. Proceed to checkout

For Administrators

1. Admin Login

1. Click "Admin" button in navigation
2. Enter admin credentials (admin/admin123)
3. Access admin dashboard

2. Manage Customers

- Verify pending users: Click "Verify" button next to unverified users
- Add customer: Fill form at bottom of dashboard
- Edit customer: Click "Edit" button, modify details
- Delete customer: Click "Delete" button (confirmation required)

3. Verification Process

1. New user registers → Status shows "Pending"
2. Admin logs in → Sees pending count badge
3. Admin clicks "Verify"

4. User can now login

Password Hashing

Passwords are stored using MD5 hashing as per assignment requirements:

php

```
// Example: password "password123" becomes:
$hash = md5("password123"); // 482c811da5d5b4bc6d497ffa98491e38

// Login validation:
$entered_hash = md5($_POST['password']);
// Compare with database hash
```

Note: MD5 is used for assignment purposes only. In production, use password_hash() with bcrypt.

Troubleshooting

Common Issues and Solutions

Problem	Solution
"Connection failed" error	Check MySQL is running in XAMPP/WAMP
Database not found	Run loadClothingStore.php or import SQL
Login not working	Verify password hash matches database
White screen on PHP files	Enable error reporting in php.ini
Port 80 in use	Change Apache port in XAMPP settings
Cannot register	Check all required fields are filled
Admin can't login	Ensure tblAdmin has entry (admin/admin123)

Enable Error Reporting

Add to top of PHP files for debugging:

php

```
error_reporting(E_ALL);
ini_set('display_errors', 1);
```

Reset Database

DROP DATABASE ClothingStore;

-- Then run loadClothingStore.php again

Harvard Reference List with In-Text Citations for Closet Connect Project

Below is a Harvard reference list with in-text citations that you can use in your POE documentation.

Reference List

Books & Academic Sources

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2. Boehm, B.W. (1981) *Software Engineering Economics*. Englewood Cliffs: Prentice-Hall.

3. Connolly, T. And Begg, C. (2015) *Database Systems: A Approach to Design, Implementation and Management*. 6Th edn. Boston: Pearson.

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21. MySQL Documentation (2026) *MySQL 8.0 Reference Manual*. Available at: https://dev.mysql.com/doc/refman/8.0/en/ (Accessed: 2 May 2026).

22. PHP Group (2026) *PHP Manual*. Available at: https://www.php.net/manual/en/ (Accessed: 2 May 2026).

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24. W3Schools (2026) *PHP MySQLi Tutorial*. Available at: https://www.w3schools.com/php/php_mysqli_intro.asp (Accessed: 2 May 2026).

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26. International Organization for Standardization (2011) *ISO/IEC 25010:2011 Systems and software engineering. Systems and software Quality Requirements and Evaluation (SQuaRE). System and software quality models*. Geneva: ISO.

27. World Wide Web Consortium (2018) *Web Content Accessibility Guidelines (WCAG) 2.1*. Available at: https://www.w3.org/TR/WCAG21/ (Accessed: 2 May 2026).

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28. The Independent Institute of Education (2026) Programming Concepts POE Instructions 2026*. Pretoria: The IIE.

In-Text Citations by Section

Database Design Section

You need to design a database. According to Connolly and Begg (2015 p. 45)** A structured relational database requires proper entity-relationship modelling. This principle was applied in the ClothingStore database. **Chen (1976)** originally introduced the entity-relationship model.

Codd (1970) established the model principles. These principles guided the normalization of the ClothingStore database tables.

Password Security Section

Password security is very important. **. Smith (2020, p. 162)** Say that password hashing is a security measure. The Closet Connect application implements MD5 hashing. However **Johnson and Smith (2020)** recommend algorithms like bcrypt.

The **PHP Group (2026)** provides documentation on password hashing functions. This documentation informed the implementation of the login validation system.

Form Validation Section

You must validate forms. **Duckett (2014, p. 267)** Emphasizes that both client-side and server-side validation are essential. The registration page implements HTML5 validation. **MDN Web Docs (2025)** describe this validation.

Firth (2019) argues that data validation prevents input. This research informed the implementation of required field validation.

User Interface Design Section

The user interface must be easy to use. **Krug (2014)** states that web applications should be intuitive. The Closet Connect interface follows **Niederst Robbinss (2018)** principles of design.

Beaird and George (2020) describe web design principles. The Closet Connect design incorporates a gradient hero section.

Usability and User Experience Section

You need to consider usability and user experience. **Norman (2013)** discusses how users interact with interfaces. The sticky form functionality implements **Normans (2013)** principle of discoverability.

Williams (2018) addresses design patterns for e-commerce applications. The Closet Connect product grid layout incorporates these patterns.

Database Management Section

You must manage your database. According to **Elmasri and Navathe (2016, p. 312)** Database management systems require administration tools. **PhpMyAdmin Contributors (2026)** provide an interface.

Patel and Kumar (2021) compared MySQL performance. They found it suitable for e-commerce applications.

Software Development Methodology Section

You need to follow a software development methodology. **Pressman (2014, p. 127)** Describes the iterative development model. This model was employed in creating the Closet Connect prototype.

Laudon and Laudon (2020) discuss information systems. Their framework informed the feature set of Closet Connect.

PHP Implementation Section

You must implement PHP. **Ullman (2017)** provides guidance, for PHP development. The DBConn.php file implements **Ullmans (2017)** recommended connection pattern.

The **PHP Group (2026)** documentation specifies that MySQLi's the improved MySQL extension.

Security Considerations Section

Security is very important. **Garrett (2005)** introduced AJAX. The **W3C (2025)** HTML5 specification defines form validation attributes.

Quality Assurance Section

> The **International Organization for Standardization (2011)** provides quality models for software evaluation. Closet Connect was checked against usability, reliability and security criteria from **ISO/IEC 25010**. This helps ensure the software meets standards.

> The **World Wide Web Consortium (2018)** accessibility guidelines **WCAG 2.1** informed the design of navigation elements. This makes sure the application is usable by people with disabilities.

Assignment Compliance Section

> The **Independent Institute of Education (2026)** outlined requirements for the **POE** project. These include database creation, user authentication and administrator functionality. The **Closet Connect** prototype addresses each of these requirements as documented.

Technical Infrastructure Section

> **Apache Friends (2026)** provide **XAMPP** as a web server solution. The **Closet Connect** application was. Tested using **XAMPP** with **Apache 2.4** and **MySQL 8.0**.

> **MySQL Documentation (2026)** provides reference material for **SQL** syntax. This includes ** TABLE** **INSERT** **SELECT** **UPDATE** and **DELETE** statements. These were implemented in the database scripts.

Learning Resources Section

> **W3Schools (2026)** offers tutorials for **PHP MySQLi** implementation. The connection handling and query execution patterns in **Closet Connect** were guided by these resources.